

COMP 2121 – Quiz 1

Time: 20 min.

Possible marks: 5

Name:

Set:

Student #:

Note: You do not need to simplify your answers unless specified otherwise.

1. Consider passwords of length 16 made up from the symbols: M, A, T, H, &, C, S

a) How many passwords contain at least one T? (0.5 marks)

All - those with no T's
 $7^{16} - 6^{16}$

b) How many passwords have exactly five T's, exactly ten H's, and one other (different) symbol? (1 mark)

$TTTTTHHHHHHHHHH*$ $\uparrow$ 5 possible symbols
 $\frac{16!}{5!10!} \times 5$

c) How many passwords with exactly five T's, exactly ten H's and one other (different) symbol have no adjacent *'s? (1 mark)

- ① Order the ten H's and * $H * H H H H H H H H H$ $\frac{11!}{10!}$ ways (or just 11)
② Insert the five T's
- choose 5 of 12 insertion points $\binom{12}{5}$ ways without repetition
③ Determine the symbol for * 5 ways

$$\binom{12}{5} \times 11 \times 5 = \frac{12!}{5!7!} \times 11 \times 5 = \frac{12!}{4!7!} \times 11$$

Formulae: $P(n, r) = \frac{n!}{(n-r)!}$

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$$

2. In how many ways can a student distribute 16 identical marbles into 5 distinct containers such that the first and last containers both have at least 2 marbles? (1 mark)

$$\boxed{2} + \boxed{} + \boxed{} + \boxed{} + \boxed{2} = \cancel{16} 12$$

① Place 2 marbles in each of first and last

② Distribute 12 remaining marbles
- choose 12 of 5 possible containers, with repetition $n=5$
 $r=12$

$$\binom{n+r-1}{r} = \binom{5+12-1}{12} = \binom{16}{12} \text{ or } \boxed{\frac{16!}{12!4!}}$$

3. What value does the variable *counter* have after the following code execution? (1.5 marks)
(Note: the *for* loops include both the start and end values).

```
counter = 300
for i = 5 to n+7 {
    counter = counter + 3
}
for j = 8 to 2n {
    counter = counter + 7
    for k = j+1 to 2n {
        counter = counter + 5
    }
}
// assume n>4
```

$$\text{Counter} = 300 + \frac{(n+3)}{1} \times 3 + \frac{(2n-7)}{2} \times 7 + \frac{(2n-7)}{2} \times 5$$

Formulae: $P(n, r) = \frac{n!}{(n-r)!}$

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$$